

Day 5 — Technical Trade-Offs & Constraints (TCD Assembly + Review)

Activity packet for your triad. Today's job: name the **top 5 trade-offs** in your architecture (TCD Section 5), build the **stakeholder sign-off matrix** (TCD Section 6), assemble the integrated TCD, peer-review it, and ship it as the sibling artifact to your PRD.

Where we are in the week

Days 1–4 produced TCD Sections 1–4 (stance, integrations + diagrams, security, SLOs). Today produces:

- **Section 5 — Top 5 trade-offs** (the section that distinguishes mature architecture documents)
- **Section 6 — Stakeholder + sign-off matrix** (who owns each constraint, what's the status)
- **The integrated TCD** — read top-to-bottom, polished, peer-reviewed, signed off

Inputs

- TCD Sections 1–4 (Days 1–4)
- The PRD it pairs with
- The "Known Trade-offs" subsection from Week 3 Day 3 — Today's trade-offs build on it
- The owner names from PRD Section 10 — they go on the sign-off matrix

What a trade-off looks like (mature vs immature)

The TCD Section 5 is the section that **separates senior architectural thinking from feature-list thinking**. A trade-off is not a list of risks; it's a structured statement of: *we considered A and B, we chose A, we accept this cost, this is what would change our minds.*

Immature	Mature
"We considered microservices but chose a monolith"	"Option A: modular monolith. Option B: extract reconcile into a service. We chose A. We accept that future service-extraction will require coordinated refactoring; we will revisit if traffic exceeds 5x rest-of-app or if a second team starts contributing reconcile features."
"Performance vs observability is a trade-off"	"Logging full event detail adds ~30ms to the hot path; we accept that until p95 nears the SLO; if the budget closes, we move to async event publication."
"We don't have time to build offline-first"	"Offline-first sync would add 8–12 weeks. We chose offline-draft + manual reconcile. Accepted cost: dispatchers without coverage during outages do paper, then re-enter. Revisit if outage frequency exceeds 1/quarter."

The pattern is always: **Option A vs Option B** → **choice** → **accepted cost** → **revisit trigger**.

The stakeholder sign-off matrix (TCD Section 6)

A constraint without an owner is a constraint nobody implements. The sign-off matrix lists every load-bearing constraint with:

- The **constraint** (link to the TCD section it's in)
- The **stakeholder** who must approve (a person, not a team)
- The **status**: proposed / discussed / approved
- The **next step** if status isn't "approved"

This section is the **action list** for Week 6 (stakeholder negotiation). Today the triad pre-loads it with everyone they'll need to talk to.

Activity 1 — Surface the Trade-Offs

Format: Triad • 35 min • Block 1

Purpose

Identify the actual trade-offs your architecture made. Most teams under-report — they describe the choice and skip the alternative.

Setup

Each triad needs TCD Sections 1–4 from this week. AI optional; use the Critique-hat prompt if you do.

The five categories of architectural trade-off

For most features, trade-offs fall into one of these:

Category	Example tension
Coupling vs independence	Module in monolith vs new service
Consistency vs availability	Strong-consistency reads vs replica reads
Latency vs durability	Sync write vs async with retry
Speed vs robustness	Ship the happy path now, harden later
Generality vs simplicity	Build the generic capability or just this case

A well-written Section 5 covers **multiple categories**. A Section 5 with 5 trade-offs all in the same category usually missed real tensions.

Triad protocol

1. **Brainstorm 8–10 candidate trade-offs** (15 min). Pull from Sections 1–4. Look for places where your TCD said "we chose X" — for each, what was the alternative?
2. **Categorize each** (5 min). Which of the 5 categories?
3. **Cull to 5 finalists** (10 min). Pick the trade-offs that are most consequential — the ones a senior architect would interrogate.
4. **Rough out the structure** (5 min). For each, jot Option A / Option B / Choice / Accepted cost / Revisit trigger.

What "good" looks like

- Trade-offs span **at least 3 categories**
- Each names a **real alternative**, not a strawman
- The "accepted cost" is concrete — not "complexity"
- The "revisit trigger" is a metric or organizational change, not "later"

Deliverable

5 trade-off finalists with sketch structure (Option A / Option B / Choice / Accepted cost / Revisit trigger), spanning at least 3 categories.

Activity 2 — Write the Top 5 Trade-Offs

Format: Triad • 40 min • Block 2

Purpose

Convert the structure from Activity 1 into the final Section 5 prose.

Setup

Each triad needs the trade-off sketches from Activity 1 and the Section 5 template. AI optional.

The trade-off template

```
### Trade-off N – <short name>
**Tension:** <category> – <one sentence framing the tension>
**Option A:** <description, 1–2 sentences>
**Option B:** <description, 1–2 sentences>
**Choice:** Option A.
**Why:** <2–3 sentences – defended in business + operational terms>
**Accepted cost:** <what we give up, concretely>
**Revisit trigger:** <metric or organizational change>
```

Worked example — FieldPulse reconcile

```
### Trade-off 1 – Module vs new service
**Tension:** Coupling vs independence.
**Option A:** Reconcile lives as a module inside the existing backend monolith.
**Option B:** Extract reconcile into a new microservice with its own database.
**Choice:** Option A.
**Why:** Only one team owns reconcile; deploy-cadence contention is zero.
    Reconcile depends on Tickets and Auth; new service buys no resilience.
    Reconcile has the same scaling curve as the rest of the app (dispatcher-shift bounded). Modular monolith is the cheaper, more reversible choice.
**Accepted cost:** Future service-extraction will require coordinated refactoring,
    especially of the Tickets boundary.
**Revisit trigger:** Reconcile traffic exceeds 5× rest-of-app, OR a second team
    begins contributing reconcile features.

### Trade-off 2 – Sync vs async write to audit
**Tension:** Latency vs durability.
**Option A:** Synchronous write to audit event store before responding to user.
**Option B:** Async publish to Kafka audit topic; consumer writes to store.
**Choice:** Option B.
**Why:** Synchronous would add 30–80ms to the reconcile latency budget, breaking the  $p95 \leq 1000\text{ms}$  SLO. Async with Kafka durability is sufficient
    for SOC 2 audit requirements. The DLQ on the consumer handles the rare
    write failures.
**Accepted cost:** Audit events may lag 1–10 seconds behind user action; a
    regulator querying within seconds of an event may see stale state.
**Revisit trigger:** A regulator or customer requires synchronous audit visibility
    within < 1 second.
```

Triad protocol

1. **Solo drafts** (15 min). Each member drafts 2 of the 5 final trade-offs.
2. **Pool and refine** (15 min). The triad polishes each — clarity, specificity.

3. **Consistency check** (10 min). Do the trade-offs **contradict** each other? If yes, that's a finding — surface it.

Deliverable

TCD Section 5 with five fully written trade-offs using the template, plus a note on any internal contradictions surfaced during the consistency check.

Activity 3 — Stakeholder Sign-Off Matrix

Format: Triad • 40 min • Block 3

Purpose

Build Section 6. Every constraint that requires another team's buy-in goes on the matrix with an owner and a status.

Setup

Each triad needs TCD Sections 1–5 from this week, PRD Section 10 (Dependencies), and any owner names surfaced earlier. AI optional.

The matrix template

```
## 6. Stakeholders + sign-off matrix
```

```
| # | Constraint (link) | Stakeholder | Status | Next step |
|---|-----|-----|-----|-----|
| 1 | Section 1 Architecture stance | <Architect name> | Proposed | Schedule 30-min walkthrough |
| 2 | Section 3 Threat model | <Security lead name> | Proposed | Send security brief, 30-min walk-through |
| 3 | Section 3 Compliance NFRs (SOC 2 audit retention) | <Compliance lead name> | Proposed | Email + 15-min sync |
| 4 | Section 4 Latency SLO | <Eng lead name> | Discussed | Confirm in next sprint review |
| 5 | Section 4 Rate-limit policy | <Platform lead name> | Proposed | Slack thread + review |
| 6 | Section 5 Trade-off 1 (module vs service) | <Architect name> | Proposed | Same walkthrough as #1 |
| 7 | Section 10 PRD: SSO scope addition | <Identity team lead> | Confirmed (Slack 2026-04-15) | None |
```

Status values:

- **Proposed:** TPM has drafted; stakeholder hasn't reacted yet
- **Discussed:** Conversation happened; no formal sign-off yet

- **Approved:** Stakeholder said yes (capture how — Slack, meeting note, email)
- **Blocked:** Stakeholder pushed back; needs different proposal

Triad protocol

1. **List every constraint that needs sign-off** (15 min). Walk through TCD Sections 1–5; for each load-bearing decision, who has to agree?
2. **Assign a real person** (10 min). Use names from PRD Section 10 + anyone new. "TBD" is not allowed.
3. **Set the next step** (10 min). For each Proposed: how will the conversation happen?
4. **Status sanity check** (5 min). Honest status — most should be Proposed.

What "good" looks like

- **At least 6 entries** — most architectural features touch this many stakeholders
- Every entry has a **real person**, not a team
- "Next step" is **specific** — "schedule a 30-min walkthrough" beats "talk to security"
- One entry probably says "Confirmed" because Week-3 Day-1 dependency work caught it; rest are Proposed

Deliverable

TCD Section 6 sign-off matrix with at least 6 named entries, each with constraint link, real-person stakeholder, honest status, and a specific next step.

Activity 4 — Integration + Cross-Review + Sign-Off

Format: Triad • 45 min + Wrap • Block 4

Purpose

Final TCD assembly. Cross-review by another triad. Internal sign-off. The TCD ships alongside the PRD.

Setup

Instructor confirms pairings. Each triad needs the full TCD Sections 1–6 draft, the integration checklist, and the Friday TCD rubric.

The integration checklist

Check	Pass criterion
Section 1 stance ↔ Section 2 diagrams	Modular monolith stance shows as one container with internal modules
Section 2 integrations ↔ Section 3 threats	Every integration boundary appears in the threat model or has an explicit "no threat" rationale
Section 3 NFRs ↔ Section 4 SLOs	Security NFRs don't conflict with latency SLOs; if they do, that's Section 5 trade-off 2
Section 4 SLOs ↔ Section 5 trade-offs	If you accepted a latency cost, it should appear in Section 5
Section 6 stakeholders ↔ Sections 1–5 constraints	Every load-bearing decision has an owner
No fortune-cookie prose	Every paragraph is specific to <i>this</i> feature

Cross-review (20 min)

Pair with another triad. The reviewer focuses on:

1. **Is each trade-off really a trade-off** or just a description of the choice?
2. **Does the stakeholder matrix have any obvious gaps?** (e.g., no security person)
3. **Are the SLOs and the trade-offs internally consistent?**

The reviewer also reads with the **Friday TCD rubric** in mind:

Dimension	Weight	Exemplary
Stance defensibility	20%	Defended in business; framework hype absent
Threat-model rigor	20%	3–5 named threats with concrete mitigations
Component clarity	15%	C4 diagram readable; externals named
SLO realism	20%	Anchored to user behavior; error budget honest
Trade-off honesty	15%	Both sides named; trigger to revisit specified
Stakeholder map	10%	Owners are people; sign-off status honest

Triad action (15 min)

Adopt / defer / reject reviewer findings. Same Week-3 discipline. Update sections.

Internal sign-off (10 min)

The triad declares the TCD **Status: Approved** (or "Approved with gaps", as Week 3). The TCD is now a sibling artifact to the PRD.

Deliverable

Signed-off TCD (Status: Approved or Approved with gaps) with adopted cross-review findings, rubric scores, and an AI provenance log covering all AI-assisted sections.

Wrap (last 15 min)

Each triad shares **one of their five trade-offs**, including the revisit trigger. The cohort gets to hear how others reasoned through architecture decisions.

End-of-week (Week 4) checkpoint

Each triad ships:

- ✓ **TCD** with Sections 1–6 complete and integrated
- ✓ **Architecture stance** with trade-off + revisit trigger (Section 1)
- ✓ **Integration map + C4 diagrams** (Section 2)
- ✓ **STRIDE threat model + revised security/compliance NFRs** (Section 3)
- ✓ **Three SLOs + latency budget + rate-limit policy + error-budget consequence** (Section 4)
- ✓ **Top 5 trade-offs** with full structure (Section 5)
- ✓ **Stakeholder sign-off matrix** with named owners (Section 6)
- ✓ AI provenance log for all AI-assisted sections

The TCD goes into Week 5 alongside the PRD. Week 5 (Technical Infrastructure & Modeling) uses the TCD's component map for data modeling, the SLOs for performance baselines, and the threat model for encryption / key-management decisions.